Freight Rate Prediction

Validation approach and December forecast

Development data and split

The labeled development file is train-test.csv. It covers posted loads from 1 January 2025 through 31 October 2025. Freight rates move with time, so the split follows the calendar instead of shuffling rows.

Fit set: 1 January through 31 August 2025, 38,477 loads. The comparison model is trained only on these rows.

Check set: 1 September through 31 October 2025, 9,523 loads. These rows are held out to measure error before the final fit.

The baseline predicts distance multiplied by the fit-set median rate per mile for that equipment type. On the check set its mean absolute error is $229.1. A LightGBM regressor trained on the same fit set has a mean absolute error of $118.6.

The final model is then retrained on every development row and saved as processed/model.pkl. That model scores all 12,000 loads in validation.csv. Predictions are written to validation_predictions.csv.

Data quality

clean.py drops duplicate load_id values. Negative weight, negative distance, and coordinates outside a valid latitude or longitude range are set to missing. Remaining numeric gaps are filled with the training median. Missing pickup, delivery, and equipment values are filled with Unknown.

validation.csv still has 165 missing weights and 249 missing market_index values. predict.py fills those with the same training medians, 31,493.5 and 1.0558, before scoring.

Model

The target is posted_rate. Features are distance, weight, market index, quote signal, pickup and delivery coordinates, month, day of week, a weekend flag, and equipment type (Dry Van, Reefer, Flatbed).

LightGBM fits a continuous dollar target when the inputs mix numbers and categories. The fitted settings are 500 trees, a learning rate of 0.05, and 31 leaves.

December 2025 predicted load rate

score.py wrote scorer_results/candidate_december.png from the completed december-chart-inputs.csv. Every row is the same load: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only the date changes, so the weekly rise and fall comes from day of week and the weekend flag. Sundays sit lower.

The predicted band is about 845*to*870. On this lane the development data has 21 Dry Van loads, with a median posted rate of about $808 and a range of about $758 to $934. The forecast sits inside that range.

